

AWS:- Cloud Computing

- **What is Cloud?**

Cloud computing is the delivery of computing services—such as servers, storage, databases, networking, software, and analytics—over the internet (“the cloud”) instead of using local servers or personal devices.

- **Types of Cloud**

There are four main types of cloud environments

1. Private Cloud
2. Public Cloud
3. Community Cloud /Multi Cloud
4. Hybrid Cloud

1. Public Cloud

Definition: A cloud environment owned and operated by third-party providers, where resources are shared among multiple customers.

Examples: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP).

Use Case: Ideal for startups, web applications, or any company that needs cost-effective and scalable infrastructure.

Advantages:

- Pay-as-you-go pricing
- High scalability and availability
- No infrastructure management required

2. Private Cloud

- **Definition:** A cloud environment dedicated to a single organization, either hosted on-premises or by a third party.

- **Use Case:** Suitable for enterprises with strict security, compliance, or regulatory requirements.

- **Advantages:**

- Greater control and customization
- Enhanced security and privacy

- Ideal for sensitive data and workloads
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3. Hybrid Cloud

- **Definition:** A combination of public and private clouds that allows data and applications to move between them.
 - **Use Case:** Ideal when you want to keep sensitive operations in a private cloud while leveraging the public cloud for less critical tasks.
 - **Advantages:**
 - Flexibility and scalability
 - Cost efficiency with control
 - Disaster recovery and business continuity
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4. Multi-Cloud

- **Definition:** The use of multiple cloud services from different providers (e.g., using both AWS and Azure).
- **Use Case:** Organizations use multi-cloud to avoid vendor lock-in, improve resilience, or optimize costs.
- **Advantages:**
 - Redundancy and risk reduction
 - Best services from different providers
 - Avoid dependency on a single vendor

Cloud Service Models

Cloud services are typically divided into **three main models**, based on what the cloud provider manages and what the customer controls:

1. IaaS (Infrastructure as a Service)

- **What it is:** Provides virtualized computing resources over the internet like servers, storage, and networking.
- **You manage:** OS, applications, data, runtime.
- **Provider manages:** Hardware, virtualization, storage, networking.
- **Examples:**
 - Amazon EC2 (AWS)
 - Microsoft Azure Virtual Machines
 - Google Compute Engine

Use Case: When you want full control over your environment but don't want to manage physical hardware.

2. PaaS (Platform as a Service)

- **What it is:** Provides a ready-to-use platform for developing, running, and managing applications without worrying about infrastructure.
- **You manage:** Application and data.
- **Provider manages:** OS, middleware, runtime, servers, storage.
- **Examples:**
 - Google App Engine
 - Microsoft Azure App Service
 - Heroku

Use Case: Best for developers who want to focus on coding and deploying applications quickly without managing infrastructure.

3. SaaS (Software as a Service)

- **What it is:** Fully functional software delivered over the internet. You just use it—everything is managed by the provider.
- **You manage:** Only user settings and usage.

- **Provider manages:** Everything (application, platform, infrastructure).
- **Examples:**
 - Gmail
 - Microsoft 365
 - Dropbox
 - Salesforce

Use Case: Ideal for end users who just need access to software via a browser or app without installation or maintenance.



Visual Summary:

	Model Managed by You	Managed by Provider	Example
IaaS	OS, apps, data	Servers, virtualization	AWS EC2
PaaS	Apps, data	OS, runtime, infrastructure	Google App Engine
SaaS	Just use the app	Everything else	Gmail, Office 365